(i) Let $u = \tan x,\ du = \sec^2 x\,dx,$ $\begin{array}{c|c|c} x & 0 & \pi/4 \\ \hline u & 0 & 1 \end{array}$

$$\int_0^{\frac{1}{4}\pi} \tan^n x \sec^2 x\,dx = \int_0^1 u^n\,du = \left[\frac{u^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1}$$

Let $u = \sec x,\ du = \sec x \tan x\,dx,$ $\begin{array}{c|c|c} x & 0 & \pi/4 \\ \hline u & 1 & \sqrt{2} \end{array}$

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} \sec^n x \tan x\,dx &= \int_1^{\sqrt{2}} u^{n-1}\,du \\ &= \left[\frac{u^n}{n}\right]_1^{\sqrt{2}} \\ &= \frac{(\sqrt{2})^n - 1}{n}\end{aligned}$$

(ii)

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} x \sec^4 x \tan x\,dx &= \left[\frac{1}{4}x\sec^4 x\right]_0^{\frac{1}{4}\pi} - \frac{1}{4}\int_0^{\frac{1}{4}\pi} \sec^4 x\,dx \\ &= \frac{\pi}{4} - \frac{1}{4}\int_0^{\frac{1}{4}\pi} (1+\tan^2 x)\sec^2 x\,dx \\ &= \frac{\pi}{4} - \frac{1}{4}\left(1 + \frac{1}{3}\right) \\ &= \frac{\pi}{4} - \frac{1}{3}\end{aligned}$$

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} x^2 \sec^2 x \tan x\,dx &= \left[\frac{1}{2}x^2\sec^2 x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} x\sec^2 x\,dx \\ &= \frac{\pi^2}{16} - \left(\left[x\tan x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \tan x\,dx\right) \\ &= \frac{\pi^2}{16} - \left(\frac{\pi}{4} - \frac{1}{2}\ln 2\right) \\ &= \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2\end{aligned}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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